

GLOBAL EARTHQUAKE DATA ANALYSIS

Project Evaluation Report

Technologies Used

Python • Pandas • Requests • MySQL • SQLAlchemy • SQL • Streamlit

Project Overview

The project collects and analyzes global earthquake data to identify useful patterns related to earthquake magnitude, depth, location, frequency, tsunami activity, and other earthquake characteristics.

Project Workflow

1	Data Collection	Collect earthquake records from the USGS earthquake API.
2	Data Processing	Convert the API response into a Pandas DataFrame.
3	Data Cleaning	Handle categorical values and missing values.
4	Feature Engineering	Create year, month, day, day-of-week and depth classification features.
5	Database Storage	Store the cleaned dataset in a MySQL table.
6	SQL Analysis	Run analytical SQL queries to answer project questions.
7	Dashboard	Display selected query results through an interactive Streamlit dashboard.

1. Data Collection

The project uses the USGS earthquake API. Python Requests is used to send API requests. The code collects data for the past five years, month by month, using a minimum earthquake magnitude of 3.

Important collected fields include:

- Earthquake ID and time
- Latitude and longitude
- Depth in kilometers
- Magnitude and magnitude type
- Place and country information
- Status and alert level
- Tsunami information
- Felt, CDI, MMI and other measurements

2. Data Cleaning

After collection, the records are converted into a Pandas DataFrame. A country column is derived from the place column. Selected categorical columns are stripped of extra spaces and converted to lowercase.

Missing values are handled using suitable statistical values. For example, median values are used for felt, gap, dmin and nst, while mean values are used for mmi, cdi and rms. Missing alert values are filled with 'green'.

3. Feature Engineering

New features are derived from the existing data. The time column is used to create year, month, day and day-of-week columns. A depth_flag column is also created: depth below 50 km is classified as 'shallow', while depth of 50 km or more is classified as 'deep'.

4. MySQL Database

The cleaned DataFrame is connected to MySQL using SQLAlchemy. The dataset is stored in a table named **earthquakes**. This allows SQL queries to be used for structured analysis.

5. SQL Analysis

The project contains multiple analytical problem statements, including:

- Top 10 strongest earthquakes.
- Top 10 deepest earthquakes.
- Earthquakes with depth below 50 km and magnitude above 7.5.
- Average magnitude by magnitude type.
- Year and month with the highest number of earthquakes.
- Day of the week with the highest earthquake count.
- Most active earthquake network.

- Tsunami-triggered earthquakes by year.
- Number of reviewed and automatic earthquakes.
- Countries with high average magnitude.
- Countries and months containing both shallow and deep earthquakes.
- Year-over-year growth rate in total earthquakes.
- Top regions based on earthquake frequency and average magnitude.
- Difference in average magnitude between tsunami and non-tsunami earthquakes.
- Events with lower data reliability based on gap and RMS.

6. Streamlit Dashboard

A Streamlit dashboard is created to make the SQL analysis interactive. The user selects a problem statement from a dropdown menu and clicks the 'Run query' button. The selected SQL query is executed using the MySQL connection, and the result is displayed as a dataframe in the dashboard.

7. Project Architecture

USGS API → Python Requests → Pandas DataFrame → Data Cleaning → Feature Engineering → MySQL → SQL Queries → Streamlit Dashboard

8. Key Skills Demonstrated

- API data collection using Python Requests.
- Data manipulation and cleaning using Pandas.
- Missing-value handling using mean and median.
- Feature engineering using date and depth information.
- MySQL database connection using SQLAlchemy.
- SQL querying, grouping, filtering, aggregation and window functions.
- Interactive dashboard development using Streamlit.

9. Conclusion

This project demonstrates an end-to-end data analysis workflow. Raw earthquake data is collected from an API, processed and cleaned with Python, transformed into useful analytical features, stored in MySQL, analyzed with SQL, and presented through an interactive Streamlit dashboard.

Project Title: Global Earthquake Data Analysis

Primary Tools: Python, Pandas, Requests, MySQL, SQL, SQLAlchemy and Streamlit